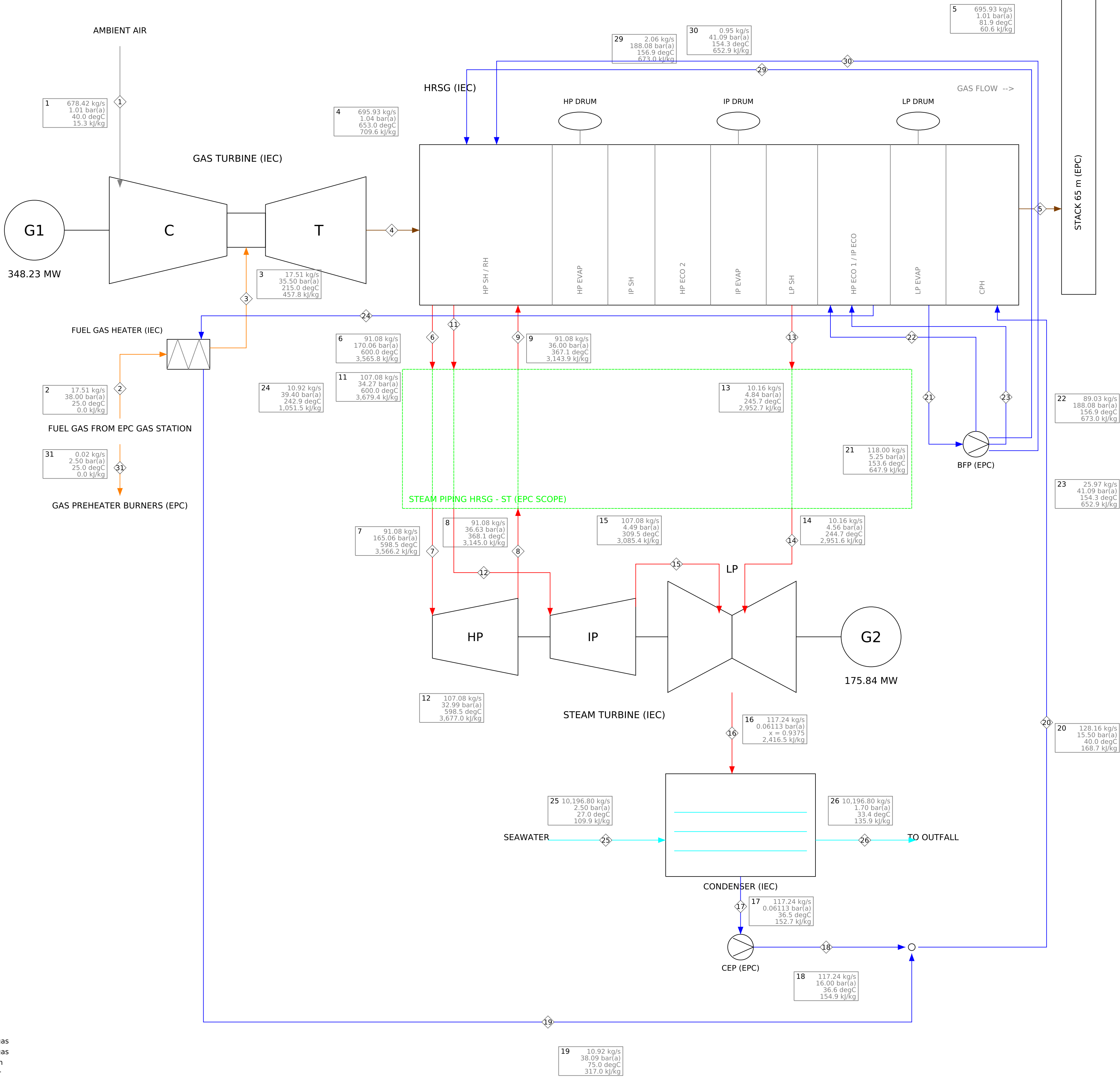


LINES
— air
— fuel gas
— flue gas
— steam
— water
— seawater
◇ stream number; data box: flow / pressure / temperature (or steam quality x) / enthalpy



STREAM TABLE - MAX-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	678.42	1.01	40.0	15.3	10.4
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	17.51	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	17.51	35.50	215.0	457.8	8.0
4	GT exhaust gas to HRSG inlet	695.93	1.04	653.0	709.6	493.8
5	Flue gas at HRSG outlet to stack	695.93	1.01	81.9	60.6	42.2
6	HP steam at HRSG outlet (HP SH outlet)	91.08	170.06	600.0	3,565.8	324.8
7	HP steam at ST main stop valve inlet	91.08	165.06	598.5	3,566.2	324.8
8	Cold reheat steam at HP turbine exhaust	91.08	36.63	368.1	3,145.0	286.5
9	Cold reheat steam at HRSG reheater inlet	91.08	36.00	367.1	3,143.9	286.4
10	IP superheated steam (joins cold reheat inside the HRSG)	15.05	36.66	327.0	3,043.8	45.8
11	Hot reheat steam at HRSG outlet	107.08	34.27	600.0	3,679.4	394.0
12	Hot reheat steam at ST IP stop/control valve inlet	107.08	32.99	598.5	3,677.0	393.7
13	LP steam at HRSG outlet	10.16	4.84	245.7	2,952.7	30.0
14	LP steam at ST LP admission	10.16	4.56	244.7	2,951.6	30.0
15	IP turbine exhaust (crossover, internal)	107.08	4.49	309.5	3,085.4	330.4
16	LP turbine exhaust to condenser (UEEP)	117.24	0.0611	x 0.938	2,416.5	283.3
17	Condensate at hotwell outlet / CEP suction	117.24	0.0611	36.5	152.7	17.9
18	Condensate at CEP discharge	117.24	16.00	36.6	154.9	18.2
19	Fuel gas heater water return to condensate	10.92	38.09	75.0	317.0	3.5
20	Condensate to HRSG condensate preheater inlet	128.16	15.50	40.0	168.7	21.6
21	Feedwater from LP drum to boiler feed pumps	118.00	5.25	153.6	647.9	76.4
22	HP feedwater at BFP discharge to HP economiser	89.03	188.08	156.9	673.0	59.9
23	IP feedwater at BFP interstage to IP economiser	25.97	41.09	154.3	652.9	17.0
24	IP economiser water to performance fuel gas heater	10.92	39.40	242.9	1,051.5	11.5
25	Circulating seawater to condenser inlet	10,196.80	2.50	27.0	109.9	1,120.4
26	Circulating seawater from condenser outlet	10,196.80	1.70	33.4	135.9	1,385.9
29	HP attemper spray water (BFP discharge)	2.06	188.08	156.9	673.0	1.4
30	Reheat attemper spray water (BFP interstage)	0.95	41.09	154.3	652.9	0.6
31	Fuel gas to the station preheater burners (water-bath heaters)	0.02	2.50	25.0	0.0	0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	348.23	MW
ST output (generator terminals)	175.84	MW
Gross output	524.07	MW
Auxiliary loads	-9.26	MW
Transformer losses (GSU, UAT, MV/LV)	-1.52	MW
NET OUTPUT (HV terminals)	513.29	MW
Heat input (LHV)	832.17	MW
Gross heat rate (LHV)	5,711	kJ/kWh
NET HEAT RATE (LHV)	5,836	kJ/kWh
Net efficiency (LHV)	61.68	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	86,136	Sm ³ /h
Circulating water flow	36,190	m ³ /h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	83.61	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.001 MW (tolerance 0.25 MW)
OK	energy balance COND: residual -0.164 MW (tolerance 1.40 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual -0.005 MW (tolerance 0.85 MW)
OK	energy balance HRSG: residual +1.830 MW vs stated casing loss 1.807 MW (tolerance 0.88 MW)
OK	energy balance PIPE: residual +0.331 MW (tolerance 1.04 MW)
OK	energy balance ST: residual -0.000 MW (tolerance 0.75 MW)
OK	gas flow 86,136 Sm ³ /h = 71.8 % of DP-QWNR-0033 (120,000 Sm ³ /h)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - MAX-NG-100
CASE	Maximum ambient (40 degC) with maximum seawater (27 degC), natural gas, base loa
DOCUMENT	ALP-HMB-MAX-NG-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	015d1cd1c6fdda54 2026-09-29
NOTE	all values calculated from the database; do not edit